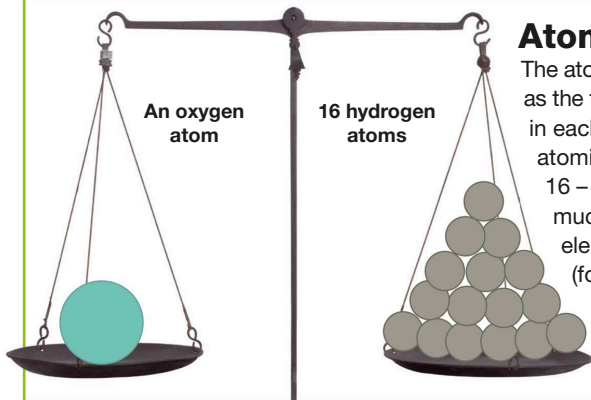


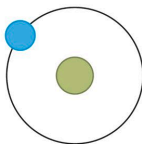


Atomic mass

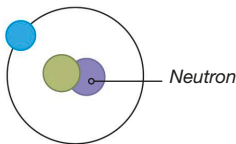
The atomic mass of an element is measured as the total number of protons and neutrons in each of its atoms. Hydrogen has an atomic mass of 1, while oxygen's is 16 – one oxygen atom weighs as much as 16 hydrogen atoms. Since an element can have different isotopes (forms), scientists use its relative atomic mass. This is the average of the atomic masses of all the isotopes of an element.

Isotopes

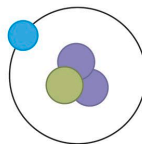
While an atom of an element has a unique number of protons and electrons, the number of neutrons in it may vary. This creates different forms of the same element, known as isotopes. For example, hydrogen has three isotopes.



Hydrogen has no neutrons.



Deuterium (an isotope of hydrogen) has 1 neutron.



Tritium (an isotope of hydrogen) has 2 neutrons.

Forming molecules

During chemical reactions, atoms make bonds with each other to form structures called molecules. Molecules of elements contain atoms of the same element, while molecules of compounds are made of atoms of different elements. For example, a water molecule forms when an oxygen atom bonds to two hydrogen atoms. This molecule is the smallest possible unit of the compound called water.

